

MYP5 Chemistry 25/26 - Equilibrium (balance) (8 weeks)

Rate of reaction
Enthalpy

Catalysts
Equilibrium

Assessment

Criterion A - end of unit assessment

Criterion B - planning a scientific investigation around the area of rate of reaction

Criterion C - analysis and evaluation of a scientific investigation around the area of rate of reaction

Criterion Div - documentings sources used in scientific investigation around the area of rate of reaction

Learning Outcome #	Curriculum	Check
1	State the definition of rate of reaction as change of concentration of reactants or products over time	
2	State the factors affecting the rate of reaction: temperature, concentration, pressure, surface area, catalyst	
3	Describe experiments to investigate the effects of changes in surface area of a solid, concentration of a solution, temperature and the use of a catalyst on the rate of a reaction	
4	Describe and Explain the effects of changes in surface area of a solid, concentration of a solution, pressure of a gas and temperature on the rate of a reaction in terms of particle collision theory	
5	State that a catalyst is a substance that increases the rate of a reaction, but is chemically unchanged at the end of the reaction	
6	State that a catalyst works by providing an alternative pathway with lower activation energy	
7	Outline that heat is released in a chemical reaction indicates an exothermic reaction ($-\Delta H$), while heat is absorbed signifies an endothermic reaction ($+\Delta H$).	
8	State that ΔH , or enthalpy change, is the amount of heat absorbed or released in a chemical reaction or physical process at constant pressure.	
9	Draw and Explain reaction profile diagrams showing ΔH and activation energy	
10	State the definition of equilibrium in a chemical reaction	
11	State that some reactions are reversible and this is indicated by the symbol $\rightleftharpoons$ in equations	
12	Define dynamic equilibrium	
13	State that a reversible reaction can reach dynamic equilibrium in a sealed container	
14	Describe the characteristics of a reaction at dynamic equilibrium: • the forward and reverse reactions occur at the same rate • the concentrations of reactants and products remain constant • that a catalyst does not affect the position of equilibrium in a reversible reaction	
15	Describe the effect of changing either temperature or pressure on the position of equilibrium in a reversible reaction: • an increase (or decrease) in temperature shifts the position of equilibrium in the direction of the endothermic (or exothermic) reaction • an increase (or decrease) in pressure shifts the position of equilibrium in the direction that produces fewer (or more) moles of gas	
16	Define Le Chatelier's principle	
17	Describe the Haber Process and the conditions for the reaction	
18	Outline economical and equilibrium considerations with respect to the Haber process	